BEFORE DAY 2 · TECH COHORT · READ ME FIRST

Come prepared: your Day 2 checklist

From Excel to Python and AI · Day 2 — AI for finance · University of Chicago

Total time: about 45 minutes
Finish **before Day 2**

You are joining on **Day 2 only** — 9:00–3:00. The room will be half you (25 tech majors) and half finance students who spent Day 1 in a full-day Python bootcamp: pandas, functions, Git, CI, and a case study on FCC broadband data. Day 2 covers how LLMs actually work, a hands-on case building and honestly evaluating an LLM news classifier for a hotel investor, and ends with an **extension sprint in mixed teams of five** — two to three finance students and two to three of you per team, each team extending its own pipeline and demoing it.

You can likely do this checklist in half the stated time. The parts not to skip are steps 3 and 4: the finance context is what your teammates bring, and knowing a little of it is what makes the teams work.

What to bring on the day

- Laptop **and charger**
- Your Google login and your GitHub login
- Chrome or Edge (Chrome is smoothest with Colab)
- Your **own API key**, obtained before the weekend through your university's AI access (see step 5) — if you can't get one, the day still works in mock mode

1 Confirm Colab works and run the setup check 10 min

The whole day runs in Google Colab — no local environment, no installs. One notebook proves your account is ready.

1. Sign in at colab.research.google.com. **Use a personal Gmail** if your university account says Colab is disabled.
2. **Open 00_setup_check in Colab** and click **Runtime → Run all**. If Colab warns the notebook *was not authored by Google*, click **Run anyway**.
3. Wait ~30 seconds and check the bottom of the notebook.

```
Loaded 236,549 rows x 12 columns
You're ready ✔  See you at 9:00.
```

✔ Expected: "You're ready" at the bottom of the notebook

If Colab is blocked on your university account, switch to a personal Gmail now — fixing it on the day costs your team time.

2 Make sure your GitHub account is ready 5 min

The finance cohort set up repositories and CI on Day 1. Sprint teams share code through GitHub, and Colab's "Save a copy in GitHub" is the move we use.

1. Sign in at github.com (or create a free account at github.com/signup with two-factor enabled).
2. Test the connection once: in any Colab notebook, click **File → Save a copy in GitHub**, authorise Colab if asked, and save to any repository you own. If that works, you are set.

3 Skim what the finance cohort built on Day 1 15 min

Day 2 builds directly on Day 1's case. You don't need to do the exercises — just read the solution notebook so the vocabulary and the data are familiar.

1. **Open 02_fcc_whitespace_SOLUTION in Colab** and read it top to bottom (don't run it — just read). It lives in a private repo: you'll get a GitHub invite after the survey — accept it first, and sign in to GitHub when Colab asks.
2. Note the shape of the work: load a 236k-row FCC broadband file, filter and group it, and find "whitespace" — census blocks underserved by current providers — for an investment screen.

- The punchline the finance cohort reached: **458 whitespace blocks, ~42,220 units**, with payback periods per county driving where to build.
- Day 2's case (`03_news_dashboard`) swaps the dataset for news headlines and the tool for an LLM — the *evaluation discipline* is the same.
- Skim the **Excel→pandas cheat sheet** (Appendix B of `docs/workshop_design.docx`) — it is your finance teammates' mental model; speak it.

4 A 10-minute primer: how a PE analyst reads the news 10 min

Day 2's case and the extension sprint are built around a question your finance teammates ask daily: “of everything published today, what actually matters to my portfolio?”

- A private-equity firm owns a handful of companies (“portfolio companies”) for 3–7 years. Analysts monitor news for **signals that change the investment case**: a competitor's acquisition, new regulation, a customer bankruptcy, management departures, input-cost shocks.
- Most news is noise. The valuable skill is **triage**: tag each story by company, theme and severity, and escalate the few that matter. That tagging job is exactly what Day 2's LLM classifier does.
- What “good” looks like to an analyst: **no missed critical stories** (recall on the bad news), tolerable false alarms, and a one-line reason they can check. A system that is 89% accurate but can't say *which* 11% it gets wrong is unusable.
- Hold that thought for the sprint rubric: every demo must name what the tool **should not be trusted for**.

Optional (10 min): 3Blue1Brown, *But what is a GPT?* — 3blue1brown.com/lessons/gpt — if you haven't seen how next-token prediction works.

5 Get an AI assistant account and know the key rules 3 min

You will use a chat assistant for debugging all day, and your own API key inside Colab for the case and the sprint.

1. Have a free account on any of: claude.ai, chatgpt.com, gemini.google.com.
2. **Bring your own API key.** Your university gives you access to AI tools and API keys — set one up **before the weekend** (it can take a few minutes and the room's wifi is not the place to discover a problem). On the day it goes in **Colab Secrets** (the key icon in the left sidebar) — **never in a code cell, never committed to GitHub**. Your key, your spend — so batch calls and prefer cheaper models for iteration. No key? Mock mode covers the whole case.

Recommended (15 min): Anthropic's prompt engineering overview — Day 2's context-engineering talk assumes nothing, but this makes the case work faster.

6 Fill in the 2-minute survey 2 min

It asks your background and GitHub username — we use it to compose the mixed teams in advance.

1. Submit it after step 2 so your GitHub username is final: **forms.gle/12uzGqjYnx3BtkD88**

The extension sprint, so you can think ahead

- **Format:** pick a direction over the working lunch, brief at 1:40, build 1:50–2:35, 2-minute demos 2:35–2:55, awards at 3:00. Teams of five, mixed finance + tech.
- **Directions** to pick from — a three-company watchlist, a cost-aware classifier, an analyst digest, a deployed dashboard — all extending the case pipeline your team just built.
- **Rubric:** does it work (30) · honest evaluation (25) · usefulness to an analyst (25) · demo quality (20). Working beats ambitious; an honest evaluation on ≥10 real items beats a perfect-looking demo.
- **Your role:** you'll likely drive the build — batching API calls, caching, a quick UI. Your finance teammates own what “useful” and “correct” mean. The best teams trade those jobs mid-way.

Stuck before the day?

1. Paste the exact error into your AI assistant with one line of context; follow its suggestion. Then search the exact error text on the web.
2. Still stuck? Email limor.gultchin@gmail.com with the step, a screenshot, and what you tried — **before Day 2**.

Doors open 8:45 on Day 2. Team assignments are on the screen when you walk in.

Done? Tick every box before Day 2

- ☐ 00_setup_check.ipynb ends with “You're ready ✔”
- ☐ Read 02_fcc_whitespace_SOLUTION top to bottom
- ☐ AI assistant account works
- ☐ Laptop charged, charger packed, both logins known
- ☐ GitHub login works; Colab → GitHub save tested
- ☐ Read the PE news-triage primer (step 4)
- ☐ Survey submitted with your GitHub username
- ☐ Skimmed the extension-sprint box above